Problem Statement #61 — Household Carbon Footprint & Target Tracker

Name: Dhriti Rajesh

SRN: PES1UG24CS907

CLASS: 5’F’

Architectural Justification

We chose Client-Server Architecture for the Household Carbon Footprint & Target Tracker System.

Architectural Choice

The system is structured around a central server that hosts the Emissions & Target Manager and Analytics & Reporting components, with lightweight clients (Resident Client UI, Coach Dashboard) that consume services over well-defined interfaces. All emissions calculations, target-progress tracking, data aggregation, and persistence are handled server-side, while the clients are responsible only for data entry and presentation.

Two Scenario-Related Reasons

1. Centralized data consistency: Per FR-003, every monthly log submission must immediately recompute the Resident's target-progress percentage, and per FR-004 the Sustainability Coach must view aggregated, anonymized trends across many households. A client-server model keeps one authoritative copy of emissions and target data on the server, so progress figures and coach-facing aggregates are always calculated from the same consistent source rather than reconciled across devices.

2. Thin, easily maintained clients: Residents and Coaches interact through simple client interfaces that only capture input (electricity, gas, transport readings) and render dashboards. Complex logic such as EPA conversion-factor calculations (FR-001, FR-002) and reminder scheduling (FR-005) lives entirely on the server, so client apps stay lightweight and can be updated or deployed to new platforms without touching core business logic.

Security Advantage

Because all household utility data is stored and encrypted in a single, hardened location (NFR-002: AES-256 at rest, TLS 1.2+ in transit) rather than replicated across many Resident and Coach devices, the server becomes the sole point that must be secured and audited. This centralization reduces the attack surface for sensitive occupancy and lifestyle data compared to a distributed or peer-to-peer alternative, and lets the Analytics & Reporting component enforce anonymization consistently before any data reaches a Coach's client.

Performance Benefit

Since CO2e conversion and target-progress calculations are performed once on the server rather than independently on every client, the Resident and Coach dashboards only need to fetch pre-computed results to render historical comparison graphs. This keeps client-side rendering well within the sub-200ms target set in NFR-001, even as the number of households and log entries grows, because the computational load is handled centrally and can be scaled or cached on the server rather than repeated per device.